

7.4 Exercises

- 7.1** One observation is taken on a discrete random variable X with pmf $f(x|\theta)$, where $\theta \in \{1, 2, 3\}$. Find the MLE of θ .

x	$f(x 1)$	$f(x 2)$	$f(x 3)$
0	$\frac{1}{3}$	$\frac{1}{4}$	0
1	$\frac{1}{3}$	$\frac{1}{4}$	0
2	0	$\frac{1}{4}$	$\frac{1}{4}$
3	$\frac{1}{6}$	$\frac{1}{4}$	$\frac{1}{2}$
4	$\frac{1}{6}$	0	$\frac{1}{4}$

- 7.2** Let $X_1, \dots, X_n$ be a random sample from a $\text{gamma}(\alpha, \beta)$ population.

- Find the MLE of β , assuming α is known.
- If α and β are both unknown, there is no explicit formula for the MLEs of α and β , but the maximum can be found numerically. The result in part (a) can be used to reduce the problem to the maximization of a univariate function. Find the MLEs for α and β for the data in Exercise 7.10(c).

- 7.3** Given a random sample $X_1, \dots, X_n$ from a population with pdf $f(x|\theta)$, show that maximizing the likelihood function, $L(\theta|\mathbf{x})$, as a function of θ is equivalent to maximizing $\log L(\theta|\mathbf{x})$.

- 7.4** Prove the assertion in Example 7.2.8. That is, prove that $\hat{\theta}$ given there is the MLE when the range of θ is restricted to the positive axis.

- 7.5** Consider estimating the binomial parameter k as in Example 7.2.9.

- Prove the assertion that the integer $\hat{k}$ that satisfies the inequalities and is the MLE is the largest integer less than or equal to $1/\bar{z}$.
- Let $p = \frac{1}{2}$, $n = 4$, and $X_1 = 0$, $X_2 = 20$, $X_3 = 1$, and $X_4 = 19$. What is $\hat{k}$?

- 7.6** Let $X_1, \dots, X_n$ be a random sample from the pdf

$$f(x|\theta) = \theta x^{-2}, \quad 0 < \theta \leq x < \infty.$$

- What is a sufficient statistic for θ ?
- Find the MLE of θ .
- Find the method of moments estimator of θ .

- 7.7** Let $X_1, \dots, X_n$ be iid with one of two pdfs. If $\theta = 0$, then

$$f(x|\theta) = \begin{cases} 1 & \text{if } 0 < x < 1 \\ 0 & \text{otherwise,} \end{cases}$$

while if $\theta = 1$, then

$$f(x|\theta) = \begin{cases} 1/(2\sqrt{x}) & \text{if } 0 < x < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find the MLE of θ .

7.8 One observation, X , is taken from a $n(0, \sigma^2)$ population.

- (a) Find an unbiased estimator of σ^2 .
- (b) Find the MLE of σ .
- (c) Discuss how the method of moments estimator of σ might be found.

7.9 Let $X_1, \dots, X_n$ be iid with pdf

$$f(x|\theta) = \frac{1}{\theta}, \quad 0 \leq x \leq \theta, \quad \theta > 0.$$

Estimate θ using both the method of moments and maximum likelihood. Calculate the means and variances of the two estimators. Which one should be preferred and why?

7.10 The independent random variables $X_1, \dots, X_n$ have the common distribution

$$P(X_i \leq x|\alpha, \beta) = \begin{cases} 0 & \text{if } x < 0 \\ (x/\beta)^\alpha & \text{if } 0 \leq x \leq \beta \\ 1 & \text{if } x > \beta, \end{cases}$$

where the parameters α and β are positive.

- (a) Find a two-dimensional sufficient statistic for (α, β) .
- (b) Find the MLEs of α and β .
- (c) The length (in millimeters) of cuckoos' eggs found in hedge sparrow nests can be modeled with this distribution. For the data

22.0, 23.9, 20.9, 23.8, 25.0, 24.0, 21.7, 23.8, 22.8, 23.1, 23.1, 23.5, 23.0, 23.0,

find the MLEs of α and β .

7.11 Let $X_1, \dots, X_n$ be iid with pdf

$$f(x|\theta) = \theta x^{\theta-1}, \quad 0 \leq x \leq 1, \quad 0 < \theta < \infty.$$

- (a) Find the MLE of θ , and show that its variance $\rightarrow 0$ as $n \rightarrow \infty$.
- (b) Find the method of moments estimator of θ .

7.12 Let $X_1, \dots, X_n$ be a random sample from a population with pmf

$$P_\theta(X = x) = \theta^x (1 - \theta)^{1-x}, \quad x = 0 \text{ or } 1, \quad 0 \leq \theta \leq \frac{1}{2}.$$

- (a) Find the method of moments estimator and MLE of θ .
- (b) Find the mean squared errors of each of the estimators.
- (c) Which estimator is preferred? Justify your choice.

7.13 Let $X_1, \dots, X_n$ be a sample from a population with double exponential pdf

$$f(x|\theta) = \frac{1}{2} e^{-|x-\theta|}, \quad -\infty < x < \infty, \quad -\infty < \theta < \infty.$$

Find the MLE of θ . (*Hint:* Consider the case of even n separate from that of odd n , and find the MLE in terms of the order statistics. A complete treatment of this problem is given in Norton 1984.)

7.14 Let X and Y be independent exponential random variables, with

$$f(x|\lambda) = \frac{1}{\lambda} e^{-x/\lambda}, \quad x > 0, \quad f(y|\mu) = \frac{1}{\mu} e^{-y/\mu}, \quad y > 0.$$

We observe Z and W with

$$Z = \min(X, Y) \quad \text{and} \quad W = \begin{cases} 1 & \text{if } Z = X \\ 0 & \text{if } Z = Y. \end{cases}$$

In Exercise 4.26 the joint distribution of Z and W was obtained. Now assume that $(Z_i, W_i), i = 1, \dots, n$, are n iid observations. Find the MLEs of λ and μ .

7.15 Let $X_1, X_2, \dots, X_n$ be a sample from the *inverse Gaussian* pdf,

$$f(x|\mu, \lambda) = \left(\frac{\lambda}{2\pi x^3}\right)^{1/2} \exp\{-\lambda(x - \mu)^2/(2\mu^2 x)\}, \quad x > 0.$$

(a) Show that the MLEs of μ and λ are

$$\hat{\mu}_n = \bar{X} \quad \text{and} \quad \hat{\lambda}_n = \frac{n}{\sum_i \frac{1}{X_i} - \frac{1}{\bar{X}}}.$$

(b) Tweedie (1957) showed that $\hat{\mu}_n$ and $\hat{\lambda}_n$ are independent, $\hat{\mu}_n$ having an inverse Gaussian distribution with parameters μ and $n\lambda$, and $n\lambda/\hat{\lambda}_n$ having a χ^2_{n-1} distribution. Schwarz and Samanta (1991) give a proof of these facts using an induction argument.

(i) Show that $\hat{\mu}_2$ has an inverse Gaussian distribution with parameters μ and 2λ , $2\lambda/\hat{\lambda}_2$ has a χ^2_1 distribution, and they are independent.

(ii) Assume the result is true for $n = k$ and that we get a new, independent observation x . Establish the induction step used by Schwarz and Samanta (1991), and transform the pdf $f(x, \hat{\mu}_k, \hat{\lambda}_k)$ to $f(x, \hat{\mu}_{k+1}, \hat{\lambda}_{k+1})$. Show that this density factors in the appropriate way and that the result of Tweedie follows.

7.16 Berger and Casella (1992) also investigate *power means*, which we have seen in Exercise 4.57. Recall that a power mean is defined as $[\frac{1}{n} \sum_{i=1}^n x_i^r]^{1/r}$. This definition can be further generalized by noting that the power function x^r can be replaced by any continuous, monotone function h , yielding the *generalized mean* $h^{-1}(\frac{1}{n} \sum_{i=1}^n h(x_i))$.

(a) The least squares problem $\min_a \sum_i (x_i - a)^2$ is sometimes solved using transformed variables, that is, solving $\min_a \sum_i [h(x_i) - h(a)]^2$. Show that the solution to this latter problem is $a = h^{-1}((1/n) \sum_i h(x_i))$.

(b) Show that the arithmetic mean is the solution to the untransformed least squares problem, the geometric mean is the solution to the problem transformed by $h(x) = \log x$, and the harmonic mean is the solution to the problem transformed by $h(x) = 1/x$.

(c) Show that if the least squares problem is transformed with the *Box-Cox Transformation* (see Exercise 11.3), then the solution is a generalized mean with $h(x) = x^\lambda$.

(d) Let $X_1, \dots, X_n$ be a sample from a $\text{lognormal}(\mu, \sigma^2)$ population. Show that the MLE of μ is the geometric mean.

(e) Suppose that $X_1, \dots, X_n$ are a sample from a one-parameter exponential family $f(x|\theta) = \exp\{\theta h(x) - H(\theta)\}g(x)$, where $h = H'$ and h is an increasing function.

(i) Show that the MLE of θ is $\hat{\theta} = h^{-1}((1/n) \sum_i h(x_i))$.

(ii) Show that two densities that satisfy $h = H'$ are the normal and the inverted gamma with pdf $f(x|\theta) = \theta x^{-2} \exp\{-\theta/x\}$ for $x > 0$, and for the normal the MLE is the arithmetic mean and for the inverted gamma it is the harmonic mean.

7.17 The Borel Paradox (Miscellanea 4.9.3) can also arise in inference problems. Suppose that X_1 and X_2 are iid exponential(θ) random variables.

- (a) If we observe only X_2 , show that the MLE of θ is $\hat{\theta} = X_2$.
- (b) Suppose that we instead observe only $Z = (X_2 - 1)/X_1$. Find the joint distribution of (X_1, Z) , and integrate out X_1 to get the likelihood function.
- (c) Suppose that $X_2 = 1$. Compare the MLEs for θ from parts (a) and (b).
- (d) Bayesian analysis is not immune to the Borel Paradox. If $\pi(\theta)$ is a prior density for θ , show that the posterior distributions, at $X_2 = 1$, are different in parts (a) and (b).

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7.18 Let $(X_1, Y_1), \dots, (X_n, Y_n)$ be iid bivariate normal random variables (pairs) where all five parameters are unknown.

- (a) Show that the method of moments estimators for $\mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2, \rho$ are $\bar{\mu}_X = \bar{x}, \bar{\mu}_Y = \bar{y}, \hat{\sigma}_X^2 = \frac{1}{n} \sum (x_i - \bar{x})^2, \hat{\sigma}_Y^2 = \frac{1}{n} \sum (y_i - \bar{y})^2, \hat{\rho} = \frac{1}{n} \sum (x_i - \bar{x})(y_i - \bar{y}) / (\hat{\sigma}_X \hat{\sigma}_Y)$.
- (b) Derive the MLEs of the unknown parameters and show that they are the same as the method of moments estimators. (One attack is to write the joint pdf as the product of a conditional and a marginal, that is, write

$$f(x, y | \mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2, \rho) = f(y | x, \mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2, \rho) f(x | \mu_X, \sigma_X^2),$$

and argue that the MLEs for μ_X and σ_X^2 are given by $\bar{x}$ and $\frac{1}{n} \sum (x_i - \bar{x})^2$. Then, turn things around to get the MLEs for μ_Y and σ_Y^2 . Finally, work with the "partially maximized" likelihood function $L(\bar{x}, \bar{y}, \hat{\sigma}_X^2, \hat{\sigma}_Y^2, \rho | \mathbf{x}, \mathbf{y})$ to get the MLE for ρ . As might be guessed, this is a difficult problem.)

7.19 Suppose that the random variables $Y_1, \dots, Y_n$ satisfy

$$Y_i = \beta x_i + \epsilon_i, \quad i = 1, \dots, n,$$

where $x_1, \dots, x_n$ are fixed constants, and $\epsilon_1, \dots, \epsilon_n$ are iid $n(0, \sigma^2)$, σ^2 unknown.

- (a) Find a two-dimensional sufficient statistic for (β, σ^2) .
- (b) Find the MLE of β , and show that it is an unbiased estimator of β .
- (c) Find the distribution of the MLE of β .

7.20 Consider $Y_1, \dots, Y_n$ as defined in Exercise 7.19.

- (a) Show that $\sum Y_i / \sum x_i$ is an unbiased estimator of β .
- (b) Calculate the exact variance of $\sum Y_i / \sum x_i$ and compare it to the variance of the MLE.

7.21 Again, let $Y_1, \dots, Y_n$ be as defined in Exercise 7.19.

- (a) Show that $[\sum (Y_i/x_i)]/n$ is also an unbiased estimator of β .
- (b) Calculate the exact variance of $[\sum (Y_i/x_i)]/n$ and compare it to the variances of the estimators in the previous two exercises.

7.22 This exercise will prove the assertions in Example 7.2.16, and more. Let $X_1, \dots, X_n$ be a random sample from a $n(\theta, \sigma^2)$ population, and suppose that the prior distribution on θ is $n(\mu, \tau^2)$. Here we assume that σ^2, μ , and τ^2 are all known.

- (a) Find the joint pdf of $\bar{X}$ and θ .
- (b) Show that $m(\bar{x} | \sigma^2, \mu, \tau^2)$, the marginal distribution of $\bar{X}$, is $n(\mu, (\sigma^2/n) + \tau^2)$.
- (c) Show that $\pi(\theta | \bar{x}, \sigma^2, \mu, \tau^2)$, the posterior distribution of θ , is normal with mean and variance given by (7.2.10).

- 7.23** If S^2 is the sample variance based on a sample of size n from a normal population, we know that $(n-1)S^2/\sigma^2$ has a χ_{n-1}^2 distribution. The conjugate prior for σ^2 is the *inverted gamma* pdf, $IG(\alpha, \beta)$, given by

$$\pi(\sigma^2) = \frac{1}{\Gamma(\alpha)\beta^\alpha} \frac{1}{(\sigma^2)^{\alpha+1}} e^{-1/(\beta\sigma^2)}, \quad 0 < \sigma^2 < \infty,$$

where α and β are positive constants. Show that the posterior distribution of σ^2 is $IG(\alpha + \frac{n-1}{2}, [\frac{(n-1)S^2}{2} + \frac{1}{\beta}]^{-1})$. Find the mean of this distribution, the Bayes estimator of σ^2 .

- 7.24** Let $X_1, \dots, X_n$ be iid $\text{Poisson}(\lambda)$, and let λ have a $\text{gamma}(\alpha, \beta)$ distribution, the conjugate family for the Poisson.

- Find the posterior distribution of λ .
- Calculate the posterior mean and variance.

- 7.25** We examine a generalization of the hierarchical (Bayes) model considered in Example 7.2.16 and Exercise 7.22. Suppose that we observe $X_1, \dots, X_n$, where

$$\begin{aligned} X_i | \theta_i &\sim n(\theta_i, \sigma^2), & i = 1, \dots, n, & \text{independent,} \\ \theta_i &\sim n(\mu, \tau^2), & i = 1, \dots, n, & \text{independent.} \end{aligned}$$

- Show that the marginal distribution of X_i is $n(\mu, \sigma^2 + \tau^2)$ and that, marginally, $X_1, \dots, X_n$ are iid. (*Empirical Bayes analysis* would use the marginal distribution of the X_i s to estimate the prior parameters μ and τ^2 . See Miscellanea 7.5.6.)
- Show, in general, that if

$$\begin{aligned} X_i | \theta_i &\sim f(x | \theta_i), & i = 1, \dots, n, & \text{independent,} \\ \theta_i &\sim \pi(\theta | \tau), & i = 1, \dots, n, & \text{independent,} \end{aligned}$$

then marginally, $X_1, \dots, X_n$ are iid.

- 7.26** In Example 7.2.16 we saw that the normal distribution is its own conjugate family. It is sometimes the case, however, that a conjugate prior does not accurately reflect prior knowledge, and a different prior is sought. Let $X_1, \dots, X_n$ be iid $n(\theta, \sigma^2)$, and let θ have a double exponential distribution, that is, $\pi(\theta) = e^{-|\theta|/a}/(2a)$, a known. Find the mean of the posterior distribution of θ .

- 7.27** Refer to Example 7.2.17.

- Show that the likelihood estimators from the complete-data likelihood (7.2.11) are given by (7.2.12).
- Show that the limit of the EM sequence in (7.2.23) satisfies (7.2.16)
- A direct solution of the original (incomplete-data) likelihood equations is possible. Show that the solution to (7.2.16) is given by

$$\hat{\beta} = \frac{\sum_{i=2}^n y_i}{\sum_{i=2}^n x_i}, \quad \hat{\tau}_1 = \frac{y_1}{\hat{\beta}}, \quad \hat{\tau}_j = \frac{x_j + y_j}{\hat{\beta} + 1}, \quad j = 2, 3, \dots, n,$$

and that this is the limit of the EM sequence in (7.2.23).

- 7.28** Use the model of Example 7.2.17 on the data in the following table adapted from Lange *et al.* (1994). These are leukemia counts and the associated populations for a number of areas in New York State.

Counts of leukemia cases

Population	3540	3560	3739	2784	2571	2729	3952	993	1908
Number of cases	3	4	1	1	3	1	2	0	2
Population	948	1172	1047	3138	5485	5554	2943	4969	4828
Number of cases	0	1	3	5	4	6	2	5	4

- (a) Fit the Poisson model to these data both to the full data set and to an “incomplete” data set where we suppose that the first population count ($x_1 = 3540$) is missing.
 (b) Suppose that instead of having an x value missing, we actually have lost a leukemia count (assume that $y_1 = 3$ is missing). Use the EM algorithm to find the MLEs in this case, and compare your answers to those of part (a).

7.29 An alternative to the model of Example 7.2.17 is the following, where we observe (Y_i, X_i) , $i = 1, 2, \dots, n$, where $Y_i \sim \text{Poisson}(m\beta\tau_i)$ and $(X_1, \dots, X_n) \sim \text{multinomial}(m; \tau)$, where $\tau = (\tau_1, \tau_2, \dots, \tau_n)$ with $\sum_{i=1}^n \tau_i = 1$. So here, for example, we assume that the population counts are multinomial allocations rather than Poisson counts. (Treat $m = \sum x_i$ as known.)

- (a) Show that the joint density of $\mathbf{Y} = (Y_1, \dots, Y_n)$ and $\mathbf{X} = (X_1, \dots, X_n)$ is

$$f(\mathbf{y}, \mathbf{x} | \beta, \tau) = \prod_{i=1}^n \frac{e^{-m\beta\tau_i} (m\beta\tau_i)^{y_i}}{y_i!} m! \frac{\tau_i^{x_i}}{x_i!}.$$

- (b) If the complete data are observed, show that the MLEs are given by

$$\hat{\beta} = \frac{\sum_{i=1}^n y_i}{\sum_{i=1}^n x_i} \quad \text{and} \quad \hat{\tau}_j = \frac{x_j + y_j}{\sum_{i=1}^n x_i + y_i}, \quad j = 1, 2, \dots, n.$$

- (c) Suppose that x_1 is missing. Use the fact that $X_1 \sim \text{binomial}(m, t_1)$ to calculate the expected complete-data log likelihood. Show that the EM sequence is given by

$$\hat{\beta}^{(r+1)} = \frac{\sum_{i=1}^n y_i}{m\hat{\tau}_1^{(r)} + \sum_{i=2}^n x_i} \quad \text{and} \quad \hat{\tau}_j^{(r+1)} = \frac{x_j + y_j}{m\hat{\tau}_1^{(r)} + \sum_{i=2}^n x_i + \sum_{i=1}^n y_i},$$

$j = 1, 2, \dots, n.$

- (d) Use this model to find the MLEs for the data in Exercise 7.28, first assuming that you have all the data and then assuming that $x_1 = 3540$ is missing.

7.30 The EM algorithm is useful in a variety of situation, and the definition of “missing data” can be stretched to accommodate many different models. Suppose that we have a mixture density $pf(x) + (1-p)g(x)$, where p is unknown. If we observe $\mathbf{X} = (X_1, \dots, X_n)$, the sample density is

$$\prod_{i=1}^n [pf(x_i) + (1-p)g(x_i)],$$

which could be difficult to deal with. (Actually, a mixture of two is not terrible, but consider what the likelihood would look like with a mixture $\sum_{i=1}^k p_i f_i(x)$ for large k .) The EM solution is to augment the observed (or incomplete) data with $\mathbf{Z} = (Z_1, \dots, Z_n)$, where Z_i tells which component of the mixture X_i came from; that is,

$$X_i | z_i = 1 \sim f(x_i) \quad \text{and} \quad X_i | z_i = 0 \sim g(x_i),$$

and $P(Z_i = 1) = p$.

- (a) Show that the joint density of $(\mathbf{X}, \mathbf{Z})$ is given by $\prod_{i=1}^n [pf(x_i)^{z_i}][(1-p)g(x_i)^{1-z_i}]$.
 (b) Show that the missing data distribution, the distribution of $Z_i|x_i, p$ is Bernoulli with success probability $pf(x_i)/(pf(x_i) + (1-p)g(x_i))$.
 (c) Calculate the expected complete-data log likelihood, and show that the EM sequence is given by

$$\hat{p}^{(r+1)} = \frac{1}{n} \sum_{i=1}^n \frac{\hat{p}^{(r)} f(x_i)}{\hat{p}^{(r)} f(x_i) + (1 - \hat{p}^{(r)}) g(x_i)}.$$

7.31 Prove Theorem 7.2.20.

- (a) Show that, using (7.2.19), we can write

$$\log L(\hat{\theta}^{(r)}|\mathbf{y}) = E[\log L(\hat{\theta}^{(r)}|\mathbf{y}, \mathbf{X})|\hat{\theta}^{(r)}, \mathbf{y}] - E[\log k(\mathbf{X}|\hat{\theta}^{(r)}, \mathbf{y})|\hat{\theta}^{(r)}, \mathbf{y}],$$

and, since $\hat{\theta}^{(r+1)}$ is a maximum, $\log L(\hat{\theta}^{(r+1)}|\mathbf{y}, \mathbf{X}) \geq E[\log L(\hat{\theta}^{(r)}|\mathbf{y}, \mathbf{X})|\hat{\theta}^{(r)}, \mathbf{y}]$. When is the inequality an equality?

- (b) Now use Jensen's inequality to show that

$$E[\log k(\mathbf{X}|\hat{\theta}^{(r+1)}, \mathbf{y})|\hat{\theta}^{(r)}, \mathbf{y}] \leq E[\log k(\mathbf{X}|\hat{\theta}^{(r)}, \mathbf{y})|\hat{\theta}^{(r)}, \mathbf{y}],$$

which together with part (a) proves the theorem.

(Hint: If f and g are densities, since $\log$ is a concave function, Jensen's inequality (4.7.7) implies

$$\int \log \left(\frac{f(x)}{g(x)} \right) g(x) dx \leq \log \left(\int \frac{f(x)}{g(x)} g(x) dx \right) = \log \left(\int f(x) dx \right) = 0.$$

By the property of logs, this in turn implies that

$$\int \log[f(x)]g(x) dx \leq \int \log[g(x)]g(x) dx.$$

7.32 The algorithm of Exercise 5.65 can be adapted to simulate (approximately) a sample from the posterior distribution using only a sample from the prior distribution. Let $X_1, \dots, X_n \sim f(x|\theta)$, where θ has prior distribution π . Generate $\theta_1, \dots, \theta_m$ from π , and calculate $q_i = L(\theta_i|\mathbf{x})/\sum_j L(\theta_j|\mathbf{x})$, where $L(\theta|\mathbf{x}) = \prod_i f(x_i|\theta)$ is the likelihood function.

- (a) Generate $\theta_1^*, \dots, \theta_r^*$, where $P(\theta^* = \theta_i) = q_i$. Show that this is a (approximate) sample from the posterior in the sense that $P(\theta^* \leq t)$ converges to $\int_{-\infty}^t \pi(\theta|\mathbf{x}) d\theta$.
 (b) Show that the estimator $\sum_{j=1}^r h(\theta_j^*)/r$ converges to $E[h(\theta)|\mathbf{x}]$, where the expectation is with respect to the posterior distribution.
 (c) Ross (1996) suggests that Rao-Blackwellization can improve the estimate in part (b). Show that for any j ,

$$E[h(\theta_j^*)|\theta_1, \dots, \theta_m] = \frac{1}{\sum_{i=1}^m L(\theta_i|\mathbf{x})} \sum_{i=1}^m h(\theta_i) L(\theta_i|\mathbf{x})$$

has the same mean and smaller variance than the estimator in part (b).

7.33 In Example 7.3.5 the MSE of the Bayes estimator, $\hat{p}_B$, of a success probability was calculated (the estimator was derived in Example 7.2.14). Show that the choice $\alpha = \beta = \sqrt{n/4}$ yields a constant MSE for $\hat{p}_B$.

7.34 Let $X_1, \dots, X_n$ be a random sample from a binomial(n, p). We want to find equivariant point estimators of p using the group described in Example 6.4.1.

- Find the class of estimators that are equivariant with respect to this group.
- Within the class of Bayes estimators of Example 7.2.14, find the estimators that are equivariant with respect to this group.
- From the equivariant Bayes estimators of part (b), find the one with the smallest MSE.

7.35 The *Pitman Estimator of Location* (see Lehmann and Casella 1998 Section 3.1, or the original paper by Pitman 1939) is given by

$$d_P(\mathbf{X}) = \frac{\int_{-\infty}^{\infty} t \prod_{i=1}^n f(x_i - t) dt}{\int_{-\infty}^{\infty} \prod_{i=1}^n f(x_i - t) dt},$$

where we observe a random sample $X_1, \dots, X_n$ from $f(x - \theta)$. Pitman showed that this estimator is the location-equivariant estimator with smallest mean squared error (that is, it minimizes (7.3.3)). The goals of this exercise are more modest.

- Show that $d_P(\mathbf{X})$ is invariant with respect to the location group of Example 7.3.6.
- Show that if $f(x - \theta)$ is $n(\theta, 1)$, then $d_P(\mathbf{X}) = \bar{X}$.
- Show that if $f(x - \theta)$ is uniform($\theta - \frac{1}{2}, \theta + \frac{1}{2}$), then $d_P(\mathbf{X}) = \frac{1}{2}(X_{(1)} + X_{(n)})$.

7.36 The *Pitman Estimator of Scale* is given by

$$d_P^r(\mathbf{X}) = \frac{\int_0^{\infty} t^{n+r-1} \prod_{i=1}^n f(tx_i) dt}{\int_0^{\infty} t^{n+2r-1} \prod_{i=1}^n f(tx_i) dt},$$

where we observe a random sample $X_1, \dots, X_n$ from $\frac{1}{\sigma} f(x/\sigma)$. Pitman showed that this estimator is the scale-equivariant estimator of σ^r with smallest scaled mean squared error (that is, it minimizes $E(d - \sigma^r)^2 / \sigma^{2r}$).

- Show that $d_P^r(\mathbf{X})$ is equivariant with respect to the scale group, that is, it satisfies

$$d_P^r(cx_1, \dots, cx_n) = c^r d_P^r(x_1, \dots, x_n),$$

for any constant $c > 0$.

- Find the Pitman scale-equivariant estimator for σ^2 if $X_1, \dots, X_n$ are iid $n(0, \sigma^2)$.
- Find the Pitman scale-equivariant estimator for β if $X_1, \dots, X_n$ are iid exponential(β).
- Find the Pitman scale-equivariant estimator for θ if $X_1, \dots, X_n$ are iid uniform($0, \theta$).

7.37 Let $X_1, \dots, X_n$ be a random sample from a population with pdf

$$f(x|\theta) = \frac{1}{2\theta}, \quad -\theta < x < \theta, \quad \theta > 0.$$

Find, if one exists, a best unbiased estimator of θ .

7.38 For each of the following distributions, let $X_1, \dots, X_n$ be a random sample. Is there a function of θ , say $g(\theta)$, for which there exists an unbiased estimator whose variance attains the Cramér–Rao Lower Bound? If so, find it. If not, show why not.

- $f(x|\theta) = \theta x^{\theta-1}$, $0 < x < 1$, $\theta > 0$
- $f(x|\theta) = \frac{\log(\theta)}{\theta-1} \theta^x$, $0 < x < 1$, $\theta > 1$

7.39 Prove Lemma 7.3.11.

7.40 Let $X_1, \dots, X_n$ be iid Bernoulli(p). Show that the variance of $\bar{X}$ attains the Cramér–Rao Lower Bound, and hence $\bar{X}$ is the best unbiased estimator of p .

- 7.41** Let $X_1, \dots, X_n$ be a random sample from a population with mean μ and variance σ^2 .
- Show that the estimator $\sum_{i=1}^n a_i X_i$ is an unbiased estimator of μ if $\sum_{i=1}^n a_i = 1$.
 - Among all unbiased estimators of this form (called *linear unbiased estimators*) find the one with minimum variance, and calculate the variance.
- 7.42** Let $W_1, \dots, W_k$ be unbiased estimators of a parameter θ with $\text{Var } W_i = \sigma_i^2$ and $\text{Cov}(W_i, W_j) = 0$ if $i \neq j$.
- Show that, of all estimators of the form $\sum a_i W_i$, where the a_i s are constant and $E(\sum a_i W_i) = \theta$, the estimator $W^* = \frac{\sum W_i / \sigma_i^2}{\sum (1/\sigma_i^2)}$ has minimum variance.
 - Show that $\text{Var } W^* = \frac{1}{\sum (1/\sigma_i^2)}$.
- 7.43** Exercise 7.42 established that the optimal weights are $q_i^* = (1/\sigma_i^2) / (\sum_j 1/\sigma_j^2)$. A result due to Tukey (see Bloch and Moses 1988) states that if $W = \sum_i q_i W_i$ is an estimator based on another sets of weights $q_i \geq 0$, $\sum_i q_i = 1$, then

$$\frac{\text{Var } W}{\text{Var } W^*} \leq \frac{1}{1 - \lambda^2},$$

where λ satisfies $(1 + \lambda)/(1 - \lambda) = b_{\max}/b_{\min}$, and $b_{\max}$ and $b_{\min}$ are the largest and smallest of $b_i = q_i/q_i^*$.

- Prove Tukey's inequality.
 - Use the inequality to assess the performance of the usual mean $\sum_i W_i/k$ as a function of $\sigma_{\max}^2/\sigma_{\min}^2$.
- 7.44** Let $X_1, \dots, X_n$ be iid $n(\theta, 1)$. Show that the best unbiased estimator of θ^2 is $\bar{X}^2 - (1/n)$. Calculate its variance (use Stein's Identity from Section 3.6), and show that it is greater than the Cramér-Rao Lower Bound.
- 7.45** Let $X_1, X_2, \dots, X_n$ be iid from a distribution with mean μ and variance σ^2 , and let S^2 be the usual unbiased estimator of σ^2 . In Example 7.3.4 we saw that, under normality, the MLE has smaller MSE than S^2 . In this exercise will explore variance estimates some more.

- Show that, for any estimator of the form aS^2 , where a is a constant,

$$\text{MSE}(aS^2) = E[aS^2 - \sigma^2]^2 = a^2 \text{Var}(S^2) + (a - 1)^2 \sigma^4.$$

- Show that

$$\text{Var}(S^2) = \frac{1}{n} \left(\kappa - \frac{n-3}{n-1} \sigma^4 \right),$$

where $\kappa = E[X - \mu]^4$ is the *kurtosis*. (You may have already done this in Exercise 5.8(b).)

- Show that, under normality, the kurtosis is $3\sigma^4$ and establish that, in this case, the estimator of the form aS^2 with the minimum MSE is $\frac{n-1}{n+1} S^2$. (Lemma 3.6.5 may be helpful.)
- If normality is not assumed, show that $\text{MSE}(aS^2)$ is minimized at

$$a = \frac{n-1}{(n+1) + \frac{(\kappa-1)(n-1)}{n}},$$

which is useless as it depends on a parameter.

(e) Show that

- (i) for distributions with $\kappa > 3$, the optimal a will satisfy $a < \frac{n-1}{n+1}$;
- (ii) for distributions with $\kappa < 3$, the optimal a will satisfy $\frac{n-1}{n+1} < a < 1$.

See Searls and Intarapanich (1990) for more details.

7.46 Let X_1, X_2 , and X_3 be a random sample of size three from a uniform($\theta, 2\theta$) distribution, where $\theta > 0$.

- (a) Find the method of moments estimator of θ .
- (b) Find the MLE, $\hat{\theta}$, and find a constant k such that $E_{\theta}(k\hat{\theta}) = \theta$.
- (c) Which of the two estimators can be improved by using sufficiency? How?
- (d) Find the method of moments estimate and the MLE of θ based on the data

1.29, .86, 1.33,

three observations of average berry sizes (in centimeters) of wine grapes.

7.47 Suppose that when the radius of a circle is measured, an error is made that has a $n(0, \sigma^2)$ distribution. If n independent measurements are made, find an unbiased estimator of the area of the circle. Is it best unbiased?

7.48 Suppose that $X_i, i = 1, \dots, n$, are iid Bernoulli(p).

- (a) Show that the variance of the MLE of p attains the Cramér–Rao Lower Bound.
- (b) For $n \geq 4$, show that the product $X_1 X_2 X_3 X_4$ is an unbiased estimator of p^4 , and use this fact to find the best unbiased estimator of p^4 .

7.49 Let $X_1, \dots, X_n$ be iid exponential(λ).

- (a) Find an unbiased estimator of λ based only on $Y = \min\{X_1, \dots, X_n\}$.
- (b) Find a better estimator than the one in part (a). Prove that it is better.
- (c) The following data are high-stress failure times (in hours) of Kevlar/epoxy spherical vessels used in a sustained pressure environment on the space shuttle:

50.1, 70.1, 137.0, 166.9, 170.5, 152.8, 80.5, 123.5, 112.6, 148.5, 160.0, 125.4.

Failure times are often modeled with the exponential distribution. Estimate the mean failure time using the estimators from parts (a) and (b).

7.50 Let $X_1, \dots, X_n$ be iid $n(\theta, \theta^2)$, $\theta > 0$. For this model both $\bar{X}$ and cS are unbiased estimators of θ , where $c = \frac{\sqrt{n-1}\Gamma((n-1)/2)}{\sqrt{2}\Gamma(n/2)}$.

- (a) Prove that for any number a the estimator $a\bar{X} + (1-a)(cS)$ is an unbiased estimator of θ .
- (b) Find the value of a that produces the estimator with minimum variance.
- (c) Show that $(\bar{X}, S^2)$ is a sufficient statistic for θ but it is not a complete sufficient statistic.

7.51 Gleser and Healy (1976) give a detailed treatment of the estimation problem in the $n(\theta, a\theta^2)$ family, where a is a known constant (of which Exercise 7.50 is a special case). We explore a small part of their results here. Again let $X_1, \dots, X_n$ be iid $n(\theta, \theta^2)$, $\theta > 0$, and let $\bar{X}$ and cS be as in Exercise 7.50. Define the class of estimators

$$\mathcal{T} = \{T: T = a_1\bar{X} + a_2(cS)\},$$

where we do not assume that $a_1 + a_2 = 1$.

- (a) Find the estimator $T \in \mathcal{T}$ that minimizes $E_{\theta}(\theta - T)^2$; call it T^* .

- (b) Show that the MSE of T^* is smaller than the MSE of the estimator derived in Exercise 7.50(b).
- (c) Show that the MSE of $T^{*+} = \max\{0, T^*\}$ is smaller than the MSE of T^* .
- (d) Would θ be classified as a location parameter or a scale parameter? Explain.
- 7.52** Let $X_1, \dots, X_n$ be iid Poisson(λ), and let $\bar{X}$ and S^2 denote the sample mean and variance, respectively. We now complete Example 7.3.8 in a different way. There we used the Cramér–Rao Bound; now we use completeness.
- (a) Prove that $\bar{X}$ is the best unbiased estimator of λ without using the Cramér–Rao Theorem.
- (b) Prove the rather remarkable identity $E(S^2|\bar{X}) = \bar{X}$, and use it to explicitly demonstrate that $\text{Var } S^2 > \text{Var } \bar{X}$.
- (c) Using completeness, can a general theorem be formulated for which the identity in part (b) is a special case?
- 7.53** Finish some of the details left out of the proof of Theorem 7.3.20. Suppose W is an unbiased estimator of $\tau(\theta)$, and U is an unbiased estimator of 0. Show that if, for some $\theta = \theta_0$, $\text{Cov}_{\theta_0}(W, U) \neq 0$, then W cannot be the best unbiased estimator of $\tau(\theta)$.
- 7.54** Consider the “Problem of the Nile” (see Exercise 6.37).
- (a) Show that T is the MLE of θ and U is ancillary, and

$$ET = \frac{\Gamma(n+1/2)\Gamma(n-1/2)}{[\Gamma(n)]^2} \theta \quad \text{and} \quad ET^2 = \frac{\Gamma(n+1)\Gamma(n-1)}{[\Gamma(n)]^2} \theta^2.$$

- (b) Let $Z_1 = (n-1)/\sum X_i$ and $Z_2 = \sum Y_i/n$. Show that both are unbiased with variances $\theta^2/(n-2)$ and θ^2/n , respectively.
- (c) Find the best unbiased estimator of the form $aZ_1 + (1-a)Z_2$, calculate its variance, and compare it to the bias-corrected MLE.
- 7.55** For each of the following pdfs, let $X_1, \dots, X_n$ be a sample from that distribution. In each case, find the best unbiased estimator of θ^r . (See Guenther 1978 for a complete discussion of this problem.)
- (a) $f(x|\theta) = \frac{1}{\theta}$, $0 < x < \theta$, $r < n$
- (b) $f(x|\theta) = e^{-(x-\theta)}$, $x > \theta$
- (c) $f(x|\theta) = \frac{e^{-x}}{e^{-\theta} - e^{-b}}$, $\theta < x < b$, b known
- 7.56** Prove the assertion made in the text preceding Example 7.3.24: If T is a complete sufficient statistic for a parameter θ , and $h(X_1, \dots, X_n)$ is any unbiased estimator of $\tau(\theta)$, then $\phi(T) = E(h(X_1, \dots, X_n)|T)$ is the best unbiased estimator of $\tau(\theta)$.
- 7.57** Let $X_1, \dots, X_{n+1}$ be iid Bernoulli(p), and define the function $h(p)$ by

$$h(p) = P\left(\sum_{i=1}^n X_i > X_{n+1} \mid p\right),$$

the probability that the first n observations exceed the $(n+1)$ st.

- (a) Show that

$$T(X_1, \dots, X_{n+1}) = \begin{cases} 1 & \text{if } \sum_{i=1}^n X_i > X_{n+1} \\ 0 & \text{otherwise} \end{cases}$$

is an unbiased estimator of $h(p)$.

- (b) Find the best unbiased estimator of $h(p)$.

7.58 Let X be an observation from the pdf

$$f(x|\theta) = \left(\frac{\theta}{2}\right)^{|x|} (1-\theta)^{1-|x|}, \quad x = -1, 0, 1; \quad 0 \leq \theta \leq 1.$$

- (a) Find the MLE of θ .
 (b) Define the estimator $T(X)$ by

$$T(X) = \begin{cases} 2 & \text{if } x = 1 \\ 0 & \text{otherwise.} \end{cases}$$

Show that $T(X)$ is an unbiased estimator of θ .

- (c) Find a better estimator than $T(X)$ and prove that it is better.

- 7.59** Let $X_1, \dots, X_n$ be iid $n(\mu, \sigma^2)$. Find the best unbiased estimator of σ^p , where p is a known positive constant, not necessarily an integer.
7.60 Let $X_1, \dots, X_n$ be iid $\text{gamma}(\alpha, \beta)$ with α known. Find the best unbiased estimator of $1/\beta$.
7.61 Show that the log of the likelihood function for estimating σ^2 , based on observing $S^2 \sim \sigma^2 \chi_\nu^2 / \nu$, can be written in the form

$$\log L(\sigma^2 | s^2) = K_1 \frac{s^2}{\sigma^2} - K_2 \log \frac{s^2}{\sigma^2} + K_3,$$

where K_1, K_2 , and K_3 are constants, not dependent on σ^2 . Relate the above log likelihood to the loss function discussed in Example 7.3.27. See Anderson (1984a) for a discussion of this relationship.

- 7.62** Let $X_1, \dots, X_n$ be a random sample from a $n(\theta, \sigma^2)$ population, σ^2 known. Consider estimating θ using squared error loss. Let $\pi(\theta)$ be a $n(\mu, \tau^2)$ prior distribution on θ and let δ^π be the Bayes estimator of θ . Verify the following formulas for the risk function and Bayes risk.

- (a) For any constants a and b , the estimator $\delta(\mathbf{x}) = a\bar{X} + b$ has risk function

$$R(\theta, \delta) = a^2 \frac{\sigma^2}{n} + (b - (1-a)\theta)^2.$$

- (b) Let $\eta = \sigma^2 / (n\tau^2 + \sigma^2)$. The risk function for the Bayes estimator is

$$R(\theta, \delta^\pi) = (1-\eta)^2 \frac{\sigma^2}{n} + \eta^2 (\theta - \mu)^2.$$

- (c) The Bayes risk for the Bayes estimator is

$$B(\pi, \delta^\pi) = \tau^2 \eta.$$

- 7.63** Let $X \sim n(\mu, 1)$. Let δ^π be the Bayes estimator of μ for squared error loss. Compute and graph the risk functions, $R(\mu, \delta^\pi)$, for $\pi(\mu) \sim n(0, 1)$ and $\pi(\mu) \sim n(0, 10)$. Comment on how the prior affects the risk function of the Bayes estimator.
7.64 Let $X_1, \dots, X_n$ be independent random variables, where X_i has cdf $F(x|\theta_i)$. Show that, for $i = 1, \dots, n$, if $\delta_i^{\pi_i}(X_i)$ is a Bayes rule for estimating θ_i using loss $L(\theta_i, a_i)$ and prior $\pi_i(\theta_i)$, then $\delta^\pi(\mathbf{X}) = (\delta^{\pi_1}(X_1), \dots, \delta^{\pi_n}(X_n))$ is a Bayes rule for estimating $\theta = (\theta_1, \dots, \theta_n)$ using the loss $\sum_{i=1}^n L(\theta_i, a_i)$ and prior $\pi(\theta) = \prod_{i=1}^n \pi_i(\theta_i)$.

- 7.65** A loss function investigated by Zellner (1986) is the LINEX (LINEar-EXponential) loss, a loss function that can handle asymmetries in a smooth way. The LINEX loss is given by

$$L(\theta, a) = e^{c(a-\theta)} - c(a - \theta) - 1,$$

where c is a positive constant. As the constant c varies, the loss function varies from very asymmetric to almost symmetric.

- For $c = .2, .5, 1$, plot $L(\theta, a)$ as a function of $a - \theta$.
 - If $X \sim F(x|\theta)$, show that the Bayes estimator of θ , using a prior π , is given by $\delta^\pi(X) = \frac{-1}{c} \log E(e^{-c\theta}|X)$.
 - Let $X_1, \dots, X_n$ be iid $n(\theta, \sigma^2)$, where σ^2 is known, and suppose that θ has the noninformative prior $\pi(\theta) = 1$. Show that the Bayes estimator versus LINEX loss is given by $\delta^B(\bar{X}) = \bar{X} - (c\sigma^2/(2n))$.
 - Calculate the posterior expected loss for $\delta^B(\bar{X})$ and $\bar{X}$ using LINEX loss.
 - Calculate the posterior expected loss for $\delta^B(\bar{X})$ and $\bar{X}$ using squared error loss.
- 7.66** The *jackknife* is a general technique for reducing bias in an estimator (Quenouille, 1956). A one-step jackknife estimator is defined as follows. Let $X_1, \dots, X_n$ be a random sample, and let $T_n = T_n(X_1, \dots, X_n)$ be some estimator of a parameter θ . In order to “jackknife” T_n we calculate the n statistics $T_n^{(i)}$, $i = 1, \dots, n$, where $T_n^{(i)}$ is calculated just as T_n but using the $n - 1$ observations with X_i removed from the sample. The jackknife estimator of θ , denoted by $JK(T_n)$, is given by

$$JK(T_n) = nT_n - \frac{n-1}{n} \sum_{i=1}^n T_n^{(i)}.$$

(In general, $JK(T_n)$ will have a smaller bias than T_n . See Miller 1974 for a good review of the properties of the jackknife.)

Now, to be specific, let $X_1, \dots, X_n$ be iid Bernoulli(θ). The object is to estimate θ^2 .

- Show that the MLE of θ^2 , $(\sum_{i=1}^n X_i/n)^2$, is a biased estimator of θ^2 .
- Derive the one-step jackknife estimator based on the MLE.
- Show that the one-step jackknife estimator is an unbiased estimator of θ^2 . (In general, jackknifing only reduces bias. In this special case, however, it removes it entirely.)
- Is this jackknife estimator the best unbiased estimator of θ^2 ? If so, prove it. If not, find the best unbiased estimator.

7.5 Miscellanea

7.5.1 Moment Estimators and MLEs

In general, method of moments estimators are not functions of sufficient statistics; hence, they can always be improved upon by conditioning on a sufficient statistic. In the case of exponential families, however, there can be a correspondence between a modified method of moments strategy and maximum likelihood estimation. This correspondence is discussed in detail by Davidson and Solomon (1974), who also relate some interesting history.